

CONSTRUCTEDGE CONSTRUCTION & INFRASTRUCTURE

2025 Procurement Performance

Cost variance, supplier pricing, procurement method, and delivery performance

— VALIDATED DECISION BRIEF —

23 July 2026

Prepared by Neil Gasson | Week 4 Midterm 1

Executive Summary

Decision: The validated dataset is fully analysis-ready. Management attention should focus on three net-overflow projects, three suppliers with a repeat above-budget pattern, and the lowest-performing delivery categories—not on a blanket change to procurement method.

KPI	Validated Result	Definition
Transactions	400	Official analysis records
Budget	\$431,518,839.83	Validated portfolio total
PO value	\$418,104,662.98	Validated portfolio total
Net cost variance	-\$13,414,176.85	3.11% under aggregate budget
Delivered-only on-time rate	74.02%	151 of 204 Delivered orders
Average / median lead time	44.65 / 42 days	Delivered orders only

The portfolio is under budget in aggregate, but the net result masks concentrated exceptions. Quarry Park Mixed-Use Development, Calgary SE Ring Road Extension, and Deerfoot Trail Widening - North are the only projects with positive net cost variance. Houle Electric, PCL Mechanical, and Levitt-Safety Calgary meet the defined consistency test for above-budget pricing. Procurement method explains only about 0.28% of row-level variance, while delivery reliability varies materially by category and supplier.

Top 3 Recommendations

- High priority — Launch targeted commercial reviews with the suppliers that meet the consistently-above-budget rule. Estimated impact: A 50% reduction in their current net positive variance equals approximately \$1,011,428.26.
- High priority — Add pre-award variance thresholds and monthly exception reviews for the three projects with positive net cost variance. Estimated impact: A 50% reduction in these project overruns equals approximately \$221,704.30.
- Medium-High priority — Create supplier delivery scorecards and corrective-action plans for material categories and suppliers below the Delivered-only portfolio benchmark. Estimated impact: Bringing below-benchmark categories up to the portfolio rate would produce approximately 7 additional on-time deliveries across the current 204 Delivered orders.

Calculation and Inclusion Rules

All Q1–Q4 results were calculated twice: Method 1 used manual Decimal/statistical aggregation in Python; Method 2 used independent pivot/groupby logic and statistical-library calculations. Every result and rank matched. The following rules are used consistently in the workbook, report, dashboard, presentation, and website content.

Question	Formula / Method	Included	Excluded	Ranking / Interpretation
Q1	$\text{SUM}(\text{Cost_Variance_CAD});$ $\text{SUM}(\text{Cost_Variance_CAD}) \div$ $\text{SUM}(\text{Budget_Value_CAD})$	All 400 records	None	Rank net variance descending
Q2	Average variance; % POs with positive variance; total variance	All 400 records	None	>50% positive POs and positive total variance; rank average variance descending
Q3	Average, median, sample standard deviation; ANOVA and Kruskal-Wallis on row Variance_%	All 400 records and all five observed methods	Invitational—no records	Descriptive association; not causal
Q4	AVERAGE/ MEDIAN Lead_Time_Days; Yes $\div$ Delivered orders	204 Delivered records	63 Pending, 65 Cancelled, 68 Partially Delivered	On-time rate desc; average lead time asc; name asc

Sign convention: Positive Cost_Variance_CAD means the purchase order is above budget. Negative variance means it is under budget. Q1 percentages use aggregate variance divided by aggregate budget—not an average of row percentages.

Q1 — Highest Cost Overruns

Exact calculation: aggregate Budget_Value_CAD, PO_Value_CAD, and Cost_Variance_CAD by project or material category. Cost Overrun (CAD) equals the sum of Cost_Variance_CAD. Cost Overrun (%) equals aggregate Cost_Variance_CAD divided by aggregate Budget_Value_CAD. All 400 records are included regardless of delivery status.

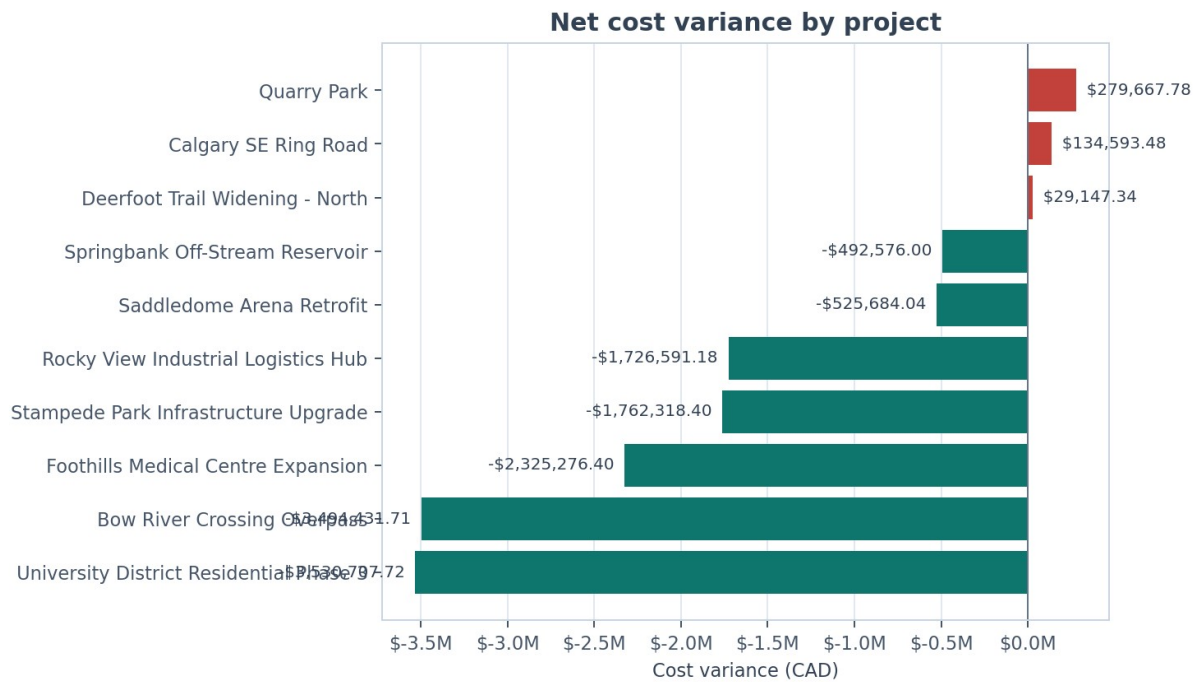


Figure 1. Net project cost variance; positive values indicate overruns.

Rank	Project	POs	Net Overrun (CAD)	Net Overrun (%)	Gross Positive Exposure
1	Quarry Park Mixed-Use Development	45	\$279,667.78	0.84%	\$1,509,812.75
2	Calgary SE Ring Road Extension	51	\$134,593.48	0.42%	\$1,755,819.27
3	Deerfoot Trail Widening - North	33	\$29,147.34	0.12%	\$1,203,606.17
4	Springbank Off-Stream Reservoir	35	-\$492,576.00	-1.66%	\$1,116,583.22
5	Saddledome Arena Retrofit	35	-\$525,684.04	-1.61%	\$869,203.87
6	Rocky View Industrial Logistics Hub	36	-\$1,726,591.18	-3.46%	\$1,877,249.43
7	Stampede Park Infrastructure Upgrade	47	-\$1,762,318.40	-2.61%	\$1,247,607.08
8	Foothills Medical Centre Expansion	45	-\$2,325,276.40	-3.54%	\$2,159,345.87
9	Bow River Crossing Overpass	46	-\$3,494,431.71	-5.48%	\$1,891,195.62
10	University District Residential Phase 3	27	-\$3,530,707.72	-11.06%	\$335,740.46

Executive interpretation: Quarry Park ranks first because its net positive variance is \$279,667.78 (0.84% of budget), the largest positive project total. Calgary SE Ring Road ranks second at \$134,593.48 (0.42%), and Deerfoot Trail ranks third at \$29,147.34 (0.12%). The other seven projects are net under budget.

Material Category Ranking

Rank	Material Category	POs	Net Overrun (CAD)	Net Overrun (%)	Gross Positive Exposure
1	Finishing	84	-\$59,993.21	-4.12%	\$36,073.62
2	Concrete & Structural	59	-\$61,347.83	-3.58%	\$43,547.84
3	Civil Works	77	-\$64,396.51	-6.62%	\$19,955.87
4	Safety & Temp Works	59	-\$441,834.52	-1.17%	\$1,511,864.52
5	MEP - Mechanical	54	-\$5,603,867.06	-3.63%	\$5,163,221.90
6	MEP - Electrical	67	-\$7,182,737.72	-3.05%	\$7,191,499.99

No material category is net over budget. Finishing ranks highest only because its net under-budget result (-\$59,993.21) is closest to zero. MEP - Electrical carries the largest gross positive exposure (\$7,191,499.99) before under-budget orders offset it; this is exposure, not a net category overrun.

Q2 — Suppliers Consistently Pricing Above Budget

Definition of “consistently”: a supplier qualifies only when more than 50% of its purchase orders have positive Cost_Variance_CAD and its total Cost_Variance_CAD is positive. All 400 records are included. Qualifiers are ranked by Average Variance (CAD) descending, with percentage of orders above budget as the tie-breaker.

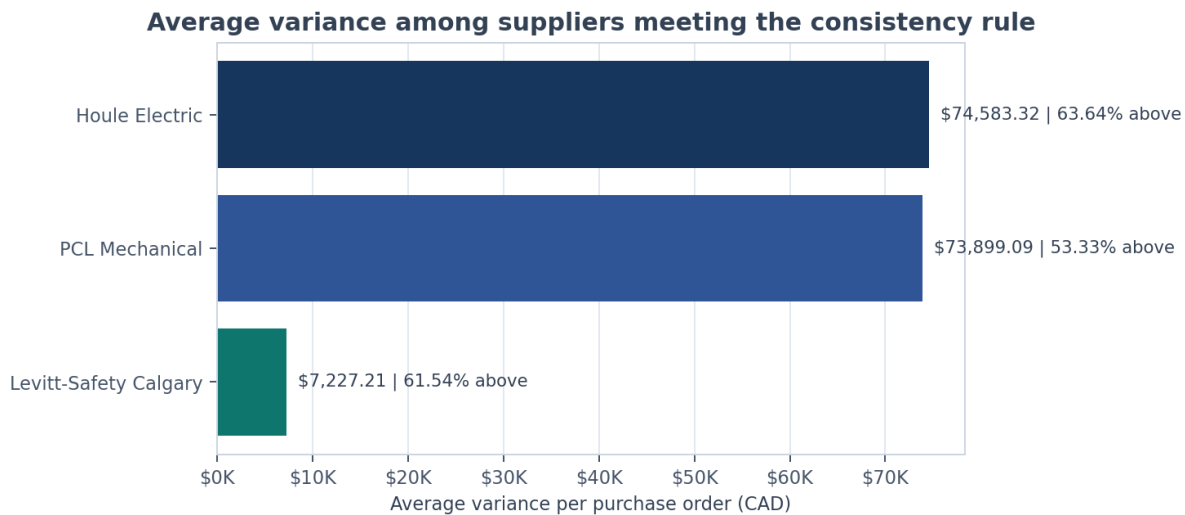


Figure 2. Average positive variance among suppliers meeting both consistency tests.

Rank	Supplier	POs	Average Variance	% Orders Above	Total Variance
1	Houle Electric	11	\$74,583.32	63.64%	\$820,416.48
2	PCL Mechanical	15	\$73,899.09	53.33%	\$1,108,486.32
3	Levitt-Safety Calgary	13	\$7,227.21	61.54%	\$93,953.71

Executive interpretation: Houle Electric ranks first with average variance of \$74,583.32 and 63.64% of orders above budget. PCL Mechanical is a close second at \$73,899.09 and 53.33%. Levitt-Safety Calgary ranks third at \$7,227.21 and 61.54%. Their combined net positive variance is \$2,022,856.51.

Complete Supplier Ranking

Rank	Supplier	POs	Avg Variance	% Above	Total Variance	Consistent?
1	Houle Electric	11	\$74,583.32	63.64%	\$820,416.48	Yes
2	PCL Mechanical	15	\$73,899.09	53.33%	\$1,108,486.32	Yes
3	Levitt-Safety Calgary	13	\$7,227.21	61.54%	\$93,953.71	Yes
—	Lafarge Canada - Calgary	13	\$378.21	30.77%	\$4,916.67	No
—	Dryvit Systems Canada	23	-\$205.30	56.52%	-\$4,722.01	No
—	Kiewit Infrastructure	18	-\$530.43	38.89%	-\$9,547.81	No
—	EllisDon Civil	15	-\$746.29	33.33%	-\$11,194.36	No
—	Centura Tile Calgary	25	-\$534.57	48.00%	-\$13,364.18	No
—	Tremco Roofing	18	-\$946.14	38.89%	-\$17,030.58	No

Rank	Supplier	POs	Avg Variance	% Above	Total Variance	Consistent?
—	Ledcor Civil Alberta	25	-\$792.56	48.00%	-\$19,813.88	No
—	Evrz Steel Calgary	16	-\$1,255.07	56.25%	-\$20,081.14	No
—	Alberta Precast	18	-\$1,276.56	38.89%	-\$22,978.08	No
—	Inland Concrete	12	-\$1,933.77	16.67%	-\$23,205.28	No
—	Graham Construction	19	-\$1,254.76	21.05%	-\$23,840.46	No
—	Armstrong World Industries	18	-\$1,382.02	22.22%	-\$24,876.44	No
—	BrandSafway Canada	13	-\$10,038.70	46.15%	-\$130,503.15	No
—	Sunbelt Rentals Canada	15	-\$10,937.37	33.33%	-\$164,060.54	No
—	Safway Alberta	18	-\$13,401.36	33.33%	-\$241,224.54	No
—	Ainsworth Inc.	10	-\$108,114.63	20.00%	-\$1,081,146.31	No
—	Johnson Controls Alberta	10	-\$136,640.16	30.00%	-\$1,366,401.60	No
—	Schneider Electric Canada	18	-\$78,439.36	66.67%	-\$1,411,908.53	No
—	Stuart Olson Electrical	20	-\$101,675.68	45.00%	-\$2,033,513.55	No
—	Stantec MEP	19	-\$224,463.45	36.84%	-\$4,264,805.47	No
—	Flint Energy Services	18	-\$253,207.34	22.22%	-\$4,557,732.12	No

Q3 — Does Procurement Method Affect Cost Variance?

All 400 purchase orders are included. The five observed methods are 3-Quote Comparison, LPO – Direct Award, Emergency Procurement, Tender – Open, and Standing Offer Agreement. The assignment requests Invitational, but the dataset contains no Invitational records, so no statistic can be calculated for that method.

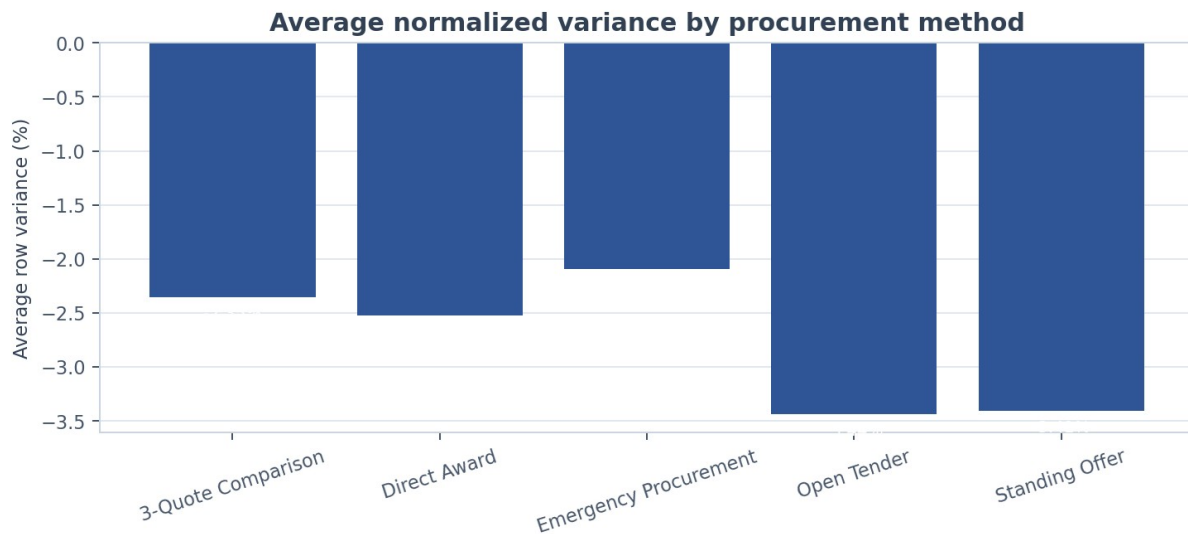


Figure 3. Every observed method has a negative average normalized variance.

Method	POs	Average Variance	Median Variance	Std. Dev.	Avg Row Variance
3-Quote Comparison	80	-\$44,467.89	-\$506.88	\$321,525.23	-2.35%
Direct Award	87	-\$21,382.80	-\$374.33	\$237,596.53	-2.52%
Emergency Procurement	76	-\$29,052.95	-\$556.82	\$210,435.48	-2.09%
Open Tender	76	-\$26,829.46	-\$682.58	\$200,547.81	-3.44%
Standing Offer	81	-\$46,288.63	-\$376.29	\$225,230.78	-3.41%

Statistical result: ANOVA on row Variance_%: $F = 0.275307$, $p = 0.893879$; eta squared = 0.002780. Kruskal-Wallis $p = 0.877840$.

Executive interpretation: procurement method does not appear to be a material driver of overruns. All five methods are net under budget, the normalized ANOVA and Kruskal-Wallis tests are far above $\alpha = 0.05$, and method explains approximately 0.28% of row-level variance. This is observational evidence and does not prove causality.

Q4 — Supplier Delivery Performance

Only records with Delivery_Status = Delivered are included. Pending, Cancelled, and Partially Delivered orders are excluded. Average Lead Time equals the mean Lead_Time_Days; Median Lead Time equals the median; On-Time Delivery Rate equals Delivered orders marked Yes divided by all Delivered orders. Rankings use on-time rate descending, average lead time ascending, then name ascending.

Metric	Result	Definition
Delivered orders	204	Denominator
Average lead time	44.65 days	Delivered only
Median lead time	42 days	Delivered only
On-time delivery	151 / 204	74.02%

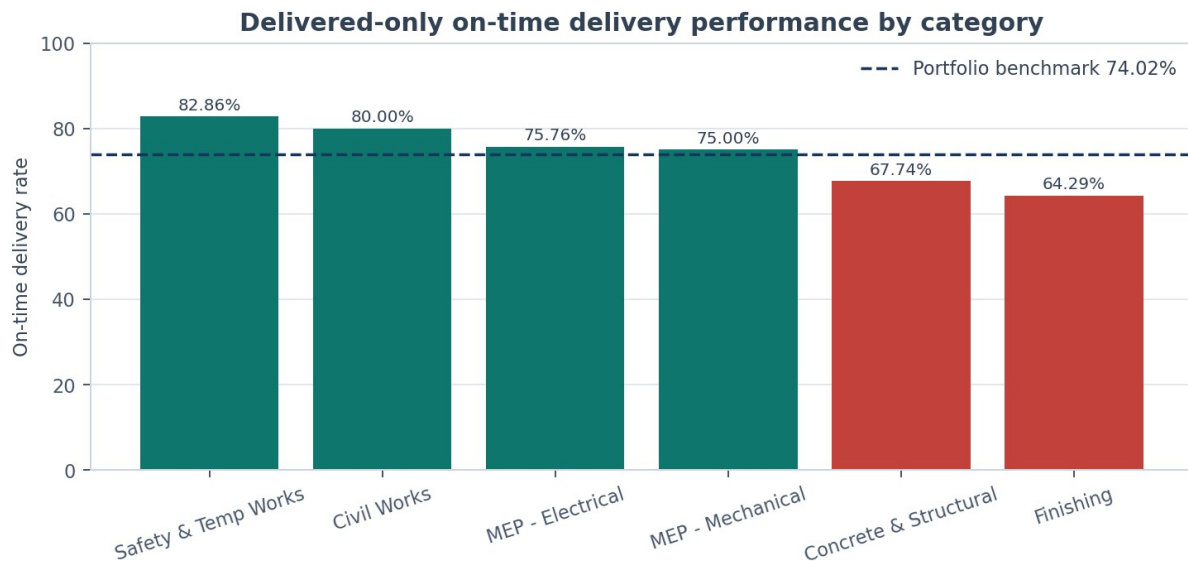


Figure 4. Material-category performance against the 74.02% Delivered-only benchmark.

Rank	Material Category	Delivered	Average Days	Median Days	On-Time Rate
1	Safety & Temp Works	35	43.94	36.00	82.86%
2	Civil Works	35	43.94	45.00	80.00%
3	MEP - Electrical	33	44.30	43.00	75.76%
4	MEP - Mechanical	28	38.93	37.50	75.00%
5	Concrete & Structural	31	48.26	46.00	67.74%
6	Finishing	42	47.24	54.00	64.29%

Executive interpretation: Safety & Temp Works ranks first at 82.86%, followed by Civil Works at 80.00%. Finishing ranks last at 64.29%, and Concrete & Structural is also below the 74.02% portfolio benchmark at 67.74%.

Supplier Delivery Ranking — Best to Worst

Rank	Supplier	Delivered	Avg Days	Median	On Time	Rate
1	Houle Electric	4	43.75	43.00	4/4	100.00%
2	Ainsworth Inc.	3	50.33	42.00	3/3	100.00%
3	Graham Construction	9	43.22	36.00	8/9	88.89%
4	Sunbelt Rentals Canada	9	45.89	43.00	8/9	88.89%
5	Levitt-Safety Calgary	9	46.11	36.00	8/9	88.89%
6	PCL Mechanical	7	36.57	41.00	6/7	85.71%
7	Inland Concrete	6	36.83	35.00	5/6	83.33%
8	BrandSafway Canada	6	49.50	46.50	5/6	83.33%
9	Armstrong World Industries	6	51.50	49.50	5/6	83.33%
10	Ledcor Civil Alberta	11	46.55	51.00	9/11	81.82%
11	Flint Energy Services	9	45.67	45.00	7/9	77.78%
12	Evrax Steel Calgary	9	48.78	46.00	7/9	77.78%
13	EllisDon Civil	8	39.25	31.50	6/8	75.00%
14	Dryvit Systems Canada	12	46.17	54.50	9/12	75.00%
15	Stantec MEP	11	36.18	33.00	8/11	72.73%
16	Safway Alberta	11	37.55	36.00	8/11	72.73%
17	Schneider Electric Canada	11	40.45	33.00	8/11	72.73%
18	Kiewit Infrastructure	7	46.14	49.00	5/7	71.43%
19	Lafarge Canada - Calgary	7	54.29	63.00	5/7	71.43%
20	Stuart Olson Electrical	9	47.89	64.00	6/9	66.67%
21	Centura Tile Calgary	17	46.65	54.00	10/17	58.82%
22	Johnson Controls Alberta	7	40.71	37.00	4/7	57.14%
23	Alberta Precast	9	50.67	49.00	4/9	44.44%
24	Tremco Roofing	7	46.86	29.00	3/7	42.86%

Houle Electric and Ainsworth Inc. each record a 100% on-time rate, but their Delivered sample sizes are only four and three orders respectively. Tremco Roofing ranks last at 42.86% (3 of 7), followed by Alberta Precast at 44.44% (4 of 9). Small samples should be considered during supplier review.

Imputation sensitivity: Leaving the two Civil Works lead times blank yields an average of 44.643564 days across 202 observations. Using the approved 45-day category median yields 44.647059 days across 204 observations—a 0.003495-day change. The median, on-time rate, category ranking, supplier ranking, and recommendations do not change.

Q5 — Top 3 Recommendations

Each recommendation below is directly supported by the validated results. Financial impact estimates are directional 50% reduction scenarios and are not additive because the supplier and project exposures can overlap.

1. Launch targeted commercial reviews with the suppliers that meet the consistently-above-budget rule.

Priority: High

Supporting evidence: 3 suppliers qualify: Houle Electric, PCL Mechanical, Levitt-Safety Calgary. Their combined net positive variance is \$2,022,856.51.

Expected business benefit: Reduces recurring supplier-level price leakage while preserving relationships with suppliers that do not show a consistent pattern.

Estimated impact: A 50% reduction in their current net positive variance equals approximately \$1,011,428.26.

2. Add pre-award variance thresholds and monthly exception reviews for the three projects with positive net cost variance.

Priority: High

Supporting evidence: 3 projects are net over budget: Quarry Park Mixed-Use Development, Calgary SE Ring Road Extension, Deerfoot Trail Widening - North. Combined net overrun is \$443,408.60.

Expected business benefit: Concentrates management attention where net overruns exist; the method analysis does not support a portfolio-wide sourcing-method change.

Estimated impact: A 50% reduction in these project overruns equals approximately \$221,704.30.

3. Create supplier delivery scorecards and corrective-action plans for material categories and suppliers below the Delivered-only portfolio benchmark.

Priority: Medium-High

Supporting evidence: The Delivered-only on-time rate is 74.02%. 2 material categories are below that benchmark.

Expected business benefit: Improves schedule reliability and makes late-delivery performance visible during supplier reviews.

Estimated impact: Bringing below-benchmark categories up to the portfolio rate would produce approximately 7 additional on-time deliveries across the current 204 Delivered orders.

Implementation sequence: 0–30 days: set supplier and project exception thresholds and assign owners. 31–60 days: complete targeted commercial reviews and launch monthly project exception reviews. 61–90 days: publish Delivered-only scorecards and require corrective-action plans for below-benchmark performance.

Limitations and Independent Verification

- The analysis is observational and based on one 2025 classroom dataset; it does not establish causation.
- Invitational procurement is absent, so Q3 cannot estimate performance for that method.
- Q4 uses Delivered orders only by instructor approval; non-Delivered statuses are intentionally excluded.
- Some supplier delivery rankings have small denominators; use them as review signals, not automatic award decisions.
- The two 45-day Civil Works lead times are instructor-approved imputations; sensitivity testing shows no ranking or recommendation change.
- Recommendation impact scenarios are directional, overlap, and should not be summed.

Question	Method 1	Method 2	Result
Q1	Manual Decimal aggregation	Pivot/groupby aggregation	MATCH
Q2	Manual Decimal aggregation	Pivot/groupby aggregation	MATCH
Q3	Manual statistics	Pivot/groupby + SciPy	MATCH
Q4	Manual Delivered-only aggregation	Pivot/groupby Delivered-only	MATCH

Final status: FULLY ANALYSIS-READY. All acceptance tests pass, both analysis methods agree for Q1–Q4, and the same validated numbers are used in every final deliverable.